

CHAPTER 14

Deterministic Operations Research

14.1. Introduction—Definition

Operations Research has been defined by the Operational Research Society of the United Kingdom as follows:

The attack of modern science on complex problems arising in the direction and management of large systems of men, machines, materials, and money in industry, business, government, and defense. The distinctive approach is to develop a scientific model of the system, incorporating measurement of factors such as chance and risk, with which to predict and compare the outcomes of alternative decisions, strategies, or controls. The purpose is to help management determine its policies and actions scientifically.

Let's examine the italicized terms more closely. First, the definition says that for operations research to be applicable, there must be systems of men, machines, materials, and money and that the techniques of O.R. are applicable to systems in industry, business, government, and defense. We can say, then, that O.R. is applicable just about anywhere there are systems that need to be managed.

The next italicized phrase, a "scientific model," is perhaps the key phrase in the definition. This implies that unless a scientific model (usually mathematical) is

developed, it cannot be called operations research. Notice that many of the techniques we have discussed in previous chapters definitely use scientific models.

The next italicized phrase says that the objectives are to “predict and compare the outcomes of alternative decisions, strategies, or controls.” This implies that any scientific model that predicts or evaluates the results of decisions or policies is operations research.

Finally, the overall objective of operations research is well stated by the last italicized phrase, “help management determine its policies and actions scientifically.” It is significant that we say to *help* management. No O.R. tool “makes” decisions. It is only an aid to the decision maker.

14.2. Similarity to Industrial Engineering

The definition of industrial engineering, as adopted by the Institute of Industrial Engineers, is as follows:

Industrial engineering is concerned with the design, improvement, and installation of integrated systems of people, materials, information, equipment, and energy. It draws upon specialized knowledge and skill in the mathematical, physical, and social sciences together with the principles and methods of engineering analysis and design to specify, predict, and evaluate the results to be obtained from such systems.

It is obvious that I.E. and O.R. have commonalities, as signified by the similarity in the definitions (note especially the italicized terms). It is the authors' opinion that operations research and industrial engineering indeed do have many of the same objectives and work on many of the same problems. The primary difference is that operations research has a higher level of theoretical and mathematical orientation, providing a major portion of the science base of industrial engineering.

Operations research carries a connotation of mathematical orientation, but industrial engineering does not restrict itself to any specific approach. Figure 14.1 illustrates the relationship between the two disciplines in terms of mathematical sophistication.

Many industrial engineers work in the area of operations research, as do

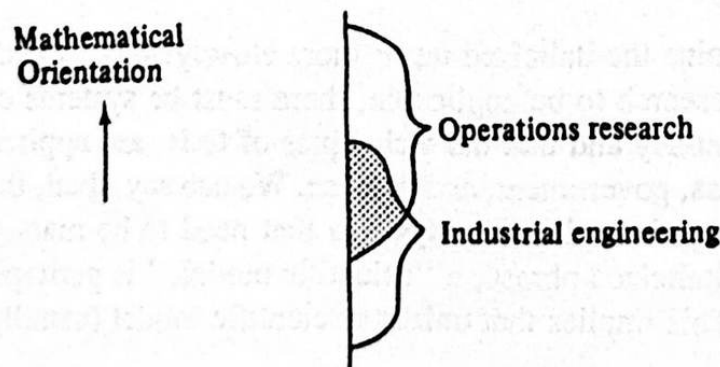


Fig. 14.1. Industrial engineering and operations research.

mathematicians, statisticians, physicists, sociologists, and others. It is significant to observe that many O.R. programs in universities are taught by industrial engineering faculty and many O.R. programs are taught by mathematics and statistics faculty. Some universities have individual programs called Operations Research that are taught by Operations Research faculty.

14.3. Nature of Operations Research

Since we have already touched upon this subject, let's explore the concept of mathematical orientation in some depth. We have said that a chief distinguishing characteristic of O.R. is the mathematical involvement. Let's look at some of the models we have already studied under industrial engineering.

14.3.1 Economic Order Quantity

In Chapter 7 we developed a mathematical model depicting the total variable cost of an inventory situation. That equation is repeated below:

$$TC = PC \frac{D}{Q} + CC \frac{Q}{2} \quad (14.1)$$

We took a first derivative, set it equal to 0, and solved for the optimum order quantity Q_0

$$Q_0 = \sqrt{\frac{(2)(PC)(D)}{CC}} \quad (14.2)$$

14.3.2 Plant Location

In Section 4.3 we developed a mathematical programming formulation of a plant location problem and stated that it could be solved by using a technique consisting of linear programming and branch and bound search (although we did not study them). That formulation is repeated below:

$$\text{Minimize } TC = \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij} + \sum_{j=1}^n k_j y_j \quad (14.3)$$

subject to

$$\sum_{i=1}^m x_{ij} \leq m y_j \quad (j = 1, \dots, n) \quad (14.4)$$

$$\sum_{j=1}^n x_{ij} = 1 \quad (i = 1, \dots, m) \quad (14.5)$$

$$y_j = 0, 1 \quad (j = 1, \dots, n) \quad (14.6)$$

$$x_{ij} \geq 0 \quad (i = 1, \dots, m \text{ and } j = 1, \dots, n) \quad (14.7)$$

14.3.3 Job Evaluation

In both point rating and factor comparison methods of job evaluation we assumed an underlying mathematical model that says that a measure of a job's total complexity can be obtained by *summing* the measures of complexity for each subfactor.

14.3.4 Quality Control

In the construction of $\bar{x}$ charts we assumed the probability distribution of sample means can be represented by the normal distribution where the normal distribution is defined (in Appendix A) as

$$f(x) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

14.3.5 Others

This discussion could be extensively continued. The transportation programming problem used in Chapters 4 and 5, the assignment problem used in Chapter 5, the computerized plant layout packages (scoring) of Chapter 4, and the traveling salesman problem and routing problems of Chapter 5 are just a few other examples of industrial engineering techniques with mathematical orientations.

Given all these formulations, would you say that they are industrial engineering or that they are operations research? Actually, they are both and would appear in the shaded area of Figure 14.1. A flow process chart might be an example of a technique that could be called industrial engineering, but not operations research (although we could argue that a flow process chart is a *scientific model* of a real-world situation). Research into new mathematical techniques could be called O.R. but not I.E., although the research is often done by industrial engineers.

14.4. Categorization of Operations Research

Let's explore the nature of operations research in an attempt to categorize the techniques. In every situation that we wish to model using an O.R. approach we encounter the problem of estimating the values of the parameters (factors such as the price of raw material or time to produce a part). Those parameters may not be constant over time. That is, they may behave as random variables or perhaps change in some predictable fashion. One approach is to forget that they are random variables and use a mathematical model that does not recognize variation. This approach is called the *deterministic* approach and is often used. Actually, all models we have seen thus far have been deterministic. If the model recognizes this random variation, the approach is called the *probabilistic* approach. We can categorize O.R. techniques

then as deterministic or probabilistic, and this is done in many I.E. and O.R. curricula in which the first course(s) explores deterministic models.

As an example (grossly oversimplified) of these two approaches, consider the inventory problem of Chapter 7. In that chapter we used a deterministic model to determine the optimum order quantity Q_0 . Now let's determine when to order if we assume that there is a lead time (delay from the time the order is issued until the material is received) involved. We shall use both a deterministic and a probabilistic approach.

14.4.1 Deterministic Approach

The model in Chapter 7 represented the inventory situation when there was zero lead time and no variation in demand. If we assume that the lead time can be specified by l and that it is deterministic (does not change), we can draw the figure for stock on hand as shown in Figure 14.2. The figure simply shows that the time to issue an order must be exactly l time units before the stock on hand runs out. This also specifies the inventory at that reorder time as

$$RQ = lD$$

where l is the lead time in units of time and D is the demand per unit of time.

14.4.2 Probabilistic Approach

Now let's assume that the demand is still constant and deterministic but that the lead time is a random variable with a mean of l . This means that we can't be sure how long after we issue an order the order will be received, but we do know that it averages l time units.

Figure 14.3 depicts the situation that might occur if we calculated the reorder quantity and time as before. We can see from Figure 14.3 that many things can occur when random variation is allowed. In the inventory situation the results usually include excess inventory at times and stockouts at other times (Figure 14.3 assumes that the customer waits and his order is filled as soon as the material arrives). This inventory model will be revisited and expanded in Chapter 15.

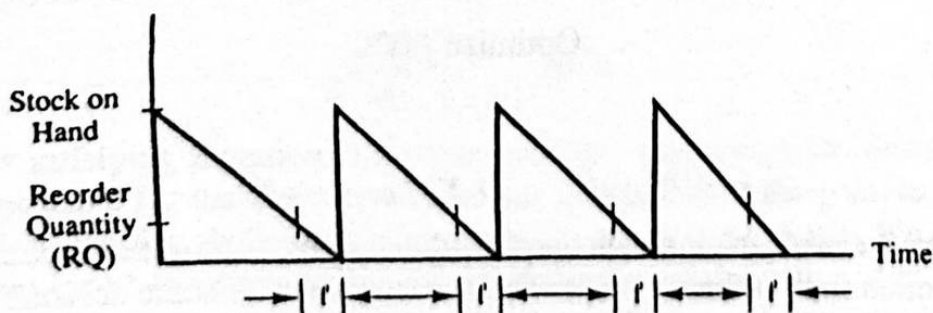


Fig. 14.2. Deterministic lead time.

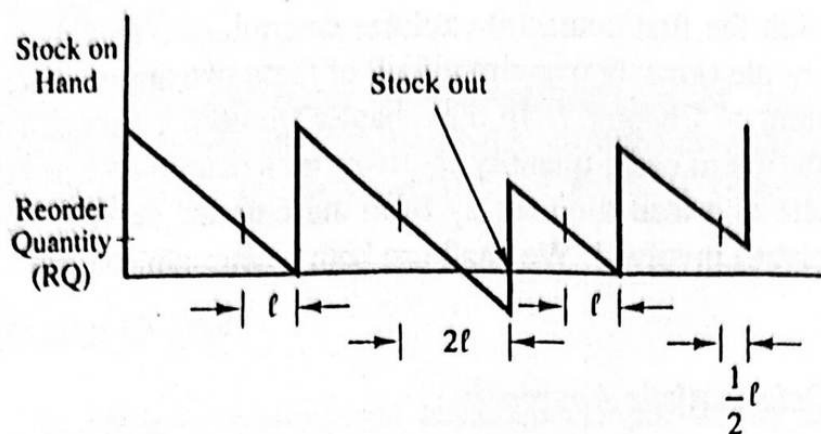


Fig. 14.3. Probabilistic lead time.

Any time random variation is allowed, the model is probabilistic; any time random variation is not allowed, the model is deterministic. The distinction will be used throughout the rest of this book. This chapter discusses some elementary and basic deterministic models. Chapter 15 briefly covers some of the probabilistic models of O.R., while Chapter 16 introduces simulation using probabilistic models.

14.5. Deterministic Operations Research

In this section and the rest of the chapter we shall cover some of the deterministic tools of operations research. We certainly cannot even mention all the areas of deterministic O.R., but we can provide a brief contact with almost all of the important ones and maybe a little insight into a few of the simpler ones. Almost all industrial engineering curricula contain several courses that cover the material presented.

14.6. Mathematical Programming

It is possible to present one general problem formulation that generalizes all deterministic operations research techniques. That problem formulation is called the *mathematical programming problem* and is presented below:

$$\text{Optimize } f(x) \quad (14.8)$$

subject to

$$g_i(x) \leq b_i \quad (i = 1, \dots, m) \quad (14.9)$$

Here, we have a mathematical function $[f(x)]$ that is to be optimized in some manner (probably minimized or maximized). The function involves some decision variables x , where x is a vector of N components $(x_1, x_2, \dots, x_N)$. There are m constraints

$$g_i(x) \leq b_i \quad (i = 1, \dots, m)$$

that restrict the choice or combination of variables so that the solution is nontrivial. (Of course, m could be zero and no constraints could be present such that an unconstrained problem like the inventory control order quantity problem results.)

There could also be other constraints. For example, some or all of the x 's could be restricted to integer quantities or just to values of 0 or 1 (as in the plant location problem formulated in Chapter 4). The decision variables could be restricted to only positive values (as is usually the case), or other special restrictions could be present. Finally, the constraints are not limited to less than or equal to inequalities. For example,

$$x_1 + x_2 \geq 5 \quad (14.10)$$

can be shown as

$$-x_1 - x_2 \leq -5 \quad (14.11)$$

and still meet the definition.

14.7. Unconstrained Optimization

The unconstrained optimization problem is one studied in calculus and is simply a mathematical programming problem where $m = 0$ (no constraints are present). The inventory control problem was just that, since the objective was to minimize the total cost equation given by

$$TC = PC \frac{D}{Q} + CC \frac{Q}{2} \quad (14.12)$$

The only constraint present is that positive quantities of Q are required, but we solved the problem without recognizing the constraint and found that there would always be a positive answer.

The unconstrained optimization problem is

$$\text{Optimize } f(x) \quad (14.13)$$

with no constraints. Calculus says that a *necessary condition* for x^* to be an optimum point of $f(x)$ is that the first derivative must be equal to zero.

$$\left. \frac{df(x)}{dx} \right|_{x=x^*} = 0 \quad (14.14)$$

A point x^* satisfying Equation (14.14) is called a *stationary point*. Since we know that it is possible for the derivative to be equal to zero and the point to be a point of inflection, the first derivative being equal to zero is not enough. We then say it is *sufficient* for x^* to be an optimum point if the second derivative at x^* is positive or negative. The point x^* is a minimum point if the second derivative is positive and a maximum point if the second derivative is negative. If the second derivative

is zero, then higher-order derivatives must be taken until the first nonzero one is found at, say, the n th derivative.

$$\left. \frac{d^n f}{dx^n} \right|_{x=x^*} \neq 0 \quad (14.15)$$

(1) If n is odd, x^* is a point of inflection.

(2) If n is even and

$$\left. \frac{d^n f}{dx^n} \right|_{x=x^*} < 0 \quad (14.16)$$

then x^* is a local maximum. If

$$\left. \frac{d^n f}{dx^n} \right|_{x=x^*} > 0 \quad (14.17)$$

then x^* is a local minimum.

As an example, consider the inventory control problem of Equation (14.12). The *necessary* condition is that

$$\left. \frac{dTC}{dQ} \right|_{Q=Q^*} = \frac{-(PC)(D)}{Q^2} + \frac{CC}{2} = 0 \quad (14.18)$$

and the *sufficient* condition for Q^* to be a minimum point is that

$$\left. \frac{d^2 TC}{dQ^2} \right|_{Q=Q^*} = \frac{2(PC)(D)}{Q^3} > 0 \quad (14.19)$$

which is greater than zero for all positive quantities of Q . Therefore, any positive Q^* that satisfies the necessary conditions of Equation (14.18) is a minimum point.

As another example, suppose that the percent of undesirable by-product in a chemical reaction is a function of time as represented by¹

$$f(t) = 3t^4 - 4t^3 + 2 \quad (14.20)$$

Suppose further that we wish to minimize this percent. The necessary condition is that

$$\frac{df(t)}{dt} = 12t^3 - 12t^2 = 0 \quad (14.21)$$

There are two points that satisfy this relationship:

$$t = 0 \quad \text{and} \quad t = 1$$

¹Extracted from William W. Claycombe and William G. Sullivan, *Foundations of Mathematical Programming* (Reston, Va.: Reston Publishing Co., Inc., 1975).

The second derivative is given by

$$\frac{d^2f(t)}{dt^2} = 36t^2 - 24t \quad (14.22)$$

At $t = 1$,

$$\frac{d^2f(t)}{dt^2} = 12 \quad (14.23)$$

Therefore, $t = 1$ is a *minimum point*.

At $t = 0$,

$$\frac{d^2f(t)}{dt^2} = 0 \quad (14.24)$$

Therefore we don't know what it is and we should take higher-order derivatives until one is nonzero.

At $n = 3$ and $t = 0$,

$$\frac{d^3f(t)}{dt^3} = 72t - 24 = -24 \quad (14.25)$$

Now, since $n = 3$ is odd, the rule states that $t = 0$ is a point of inflection.

For a geometrical interpretation of the above, consider Figures 14.4, 14.5 and 14.6. In all the figures, x^* satisfies the necessary condition that the first derivative equals zero. In Figure 14.4 the second derivative is positive at x^* ; therefore, x^* is a minimum point. Note also that the second derivative would be positive at all points. A function like this is said to be *convex* and any stationary point is a minimum point for a convex function.

In Figure 14.5 the second derivative is negative for all values of x ; therefore, any stationary point should be a maximum. There is only one stationary point and that is x^* which is a maximum point. A function whose second derivative is negative for all values of x is said to be *concave*.

Figure 14.6 is slightly different. We can see that for any $x < x^*$, the second derivative is negative; for any $x > x^*$, the second derivative is positive; and at

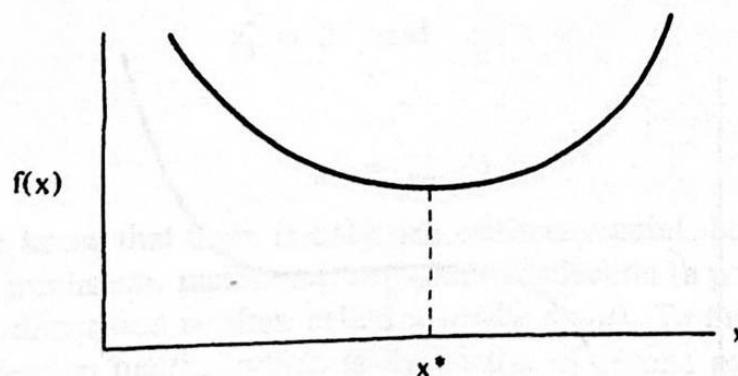


Fig. 14.4. Minimum point-single variable.

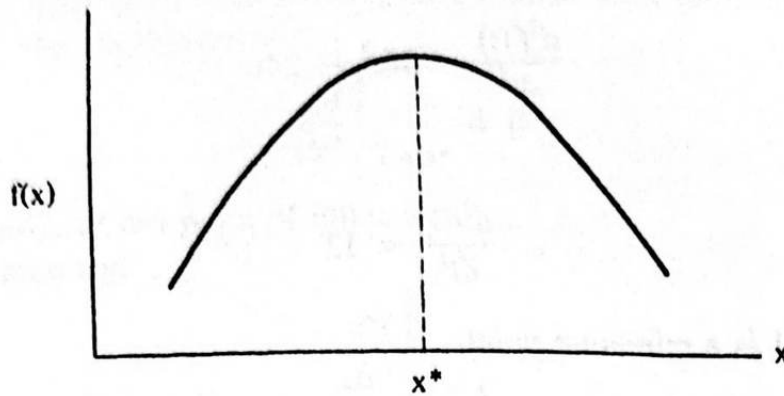


Fig. 14.5. Maximum point—single variable.

$x = x^*$, the second derivative is zero. If continued, we would find that some higher-order n th derivative, where n is odd, evaluated at x^* would be nonzero, signifying that x^* is a point of inflection, which it is.

Finally, it is possible for a function to have several minima, maxima, and points of inflection. Figure 14.7 demonstrates this possibility.

In this figure x_1^* , x_2^* , x_3^* , and x_4^* are all stationary points (first derivative = 0). Points x_1^* and x_3^* are minimum points (second derivative positive), x_2^* is a maximum point (second derivative negative), and x_4^* is a point of inflection (second derivative equal to zero and some higher-odd order derivative not equal to zero). Of course, if the function is convex (concave) for all values of x , there can only be one stationary point and that point is a unique minimum (maximum). When there is only one minimum (maximum), that minimum (maximum) is called the *global* minimum (maximum). When there is more than one minimum (maximum), the smallest of all the minima (largest of all the maxima) is also called the *global* minimum (maximum).

For functions of more than one variable, the same essential relationships hold. For example, the necessary condition for a vector

$$x^* = (x_1, x_2, \dots, x_n) \quad (14.26)$$

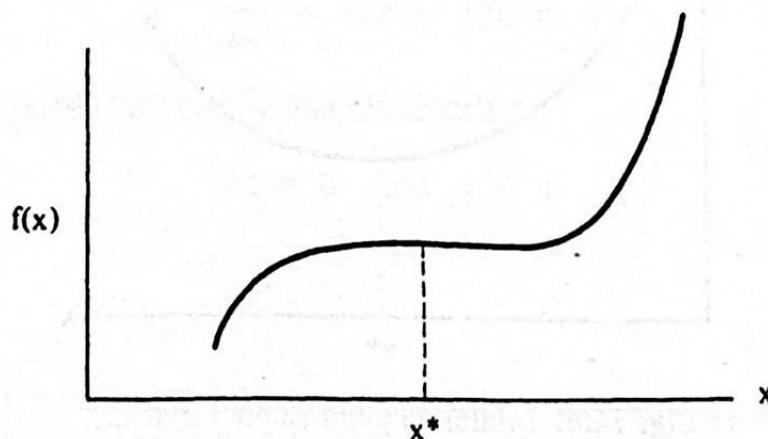


Fig. 14.6. Point of inflection—single variable.

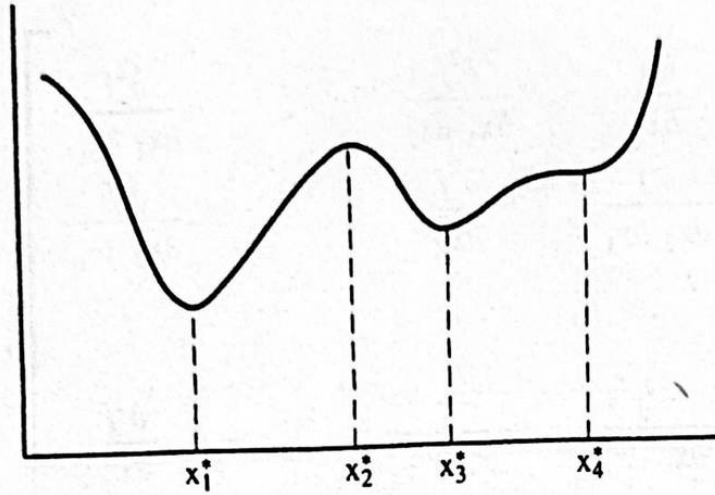


Fig. 14.7. Function with multiple stationary points.

to be a stationary point of a function $F(x)$, all elements of the vector of first partial derivatives (called the *gradient*) evaluated at x^* must be zero.

$$\nabla F(x) = \left(\frac{\partial F(x)}{\partial x_1}, \frac{\partial F(x)}{\partial x_2}, \dots, \frac{\partial F(x)}{\partial x_n} \right)^T = 0 \quad (14.27)$$

As an example, suppose that the cost of producing a product in a continuous process is a function of the amount of material added (x_1) and the temperature (x_2) as shown in

$$TC = 2x_1^2 - 2x_1x_2 + x_2^2 - 4x_2 + 5 \quad (14.28)$$

The necessary condition [Equation (14.27)] states that

$$\frac{\partial TC}{\partial x_1} = 4x_1 - 2x_2 = 0 \quad (14.29)$$

and

$$\frac{\partial TC}{\partial x_2} = -2x_1 + 2x_2 - 4 = 0 \quad (14.30)$$

Solving these two equations simultaneously yields

$$x_1^* = 2 \quad \text{and} \quad x_2^* = 4 \quad (14.31)$$

or

$$x^* = (2, 4) \quad (14.32)$$

Now, we know that there is only one stationary point, but we do not know whether it is a minimum, maximum, or point of inflection (a point of inflection in more than one dimension is often called a *saddle point*). To find this, we need to evaluate the Hessian matrix, which is the matrix of second partials as shown in Figure 14.8.

The sufficiency test involves testing this Hessian matrix for something called

$$\begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \dots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \dots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

Fig. 14.8. Hessian matrix.

positive or negative definiteness (note the similarity to the one-dimensional case). If the matrix is positive definite at a stationary point x^* , then x^* is a minimum point; if the matrix is negative definite, then x^* is a maximum point. Testing for this definiteness is beyond the scope of this book, but it involves a check of the eigenvalues of the matrix or calculation of determinants of the principle minor submatrices. As for the single variable case, we can show that if the Hessian matrix is positive (negative) definite for all values of x , then any stationary point is a minimum (maximum). The function is then said to be convex (concave).

Going back to the example problem of Equation (14.28), we can see that the Hessian matrix is

$$\begin{bmatrix} 4 & -2 \\ -2 & 2 \end{bmatrix}$$

We can show that this matrix is positive definite for all values of x . Therefore, the function is convex and any stationary point is a minimum point. The point

$$x^* = (2, 4) \quad (14.33)$$

is a minimum point.

Sometimes the methods of calculus covered above are not applicable for one of several possible reasons. The most frequently occurring is that the mathematical model of the system is not known. For example, a new chemical process may have several variables, including items such as composition, temperature, and humidity and to run the process may be extremely costly. Therefore, it is virtually impossible to run the system often enough to be able to *fit* a mathematical function to the results. It is necessary, however, to be able to determine *good* if not *optimum* values of the variables. This also occurs when a system is being *simulated* on a computer (to be covered in Chapter 16). Another problem may occur when the mathematical function can be stated, but taking derivatives, etc., would be very difficult. In situations like these the techniques of *search theory* are often used.

Search theory, as the name implies, is a *search* over the feasible domain of

the variables in an effort to determine *good* values for the variables. There are two essential search procedures—those that use derivatives and those that don't. Search procedures that use derivatives are limited in scope because derivatives aren't always available and they are more difficult to explain than the procedures that don't require derivatives. Consequently, we shall cover only that class of search procedures that does not require derivatives.

For one-dimensional problems, there are several very efficient search procedures available, for example, Bolzano, Fibonacci, paired block, and golden section. We shall cover here only the paired block procedure. Suppose that we have a production process whose total cost is completely dependent on the temperature of the process and suppose that we wish to minimize this total cost by choosing a good value for the temperature. Suppose also that it is possible to say that the temperature must lie between 50 and 150°F and our controls are capable of distinguishing only full degree differences, not fractional ones. (Minimum spacing for distinguishing between values is called *resolution*.) The paired block procedure places two observations as close together as possible (i.e., one resolution apart) in the middle of the range. In our example the process should be run at 99.5 and 100.5°F for the first two trials. Since we can only run at full degrees, we arbitrarily choose to run at 100 and 101°F, yielding the results shown in Figure 14.9 in which the height of the vertical line represents the total cost of the process at those points. Obviously, 101°F is cheaper and, therefore, most likely (definitely if the function is convex) the best value will lie between 100 and 150°F. We should then try the process at two points close to the middle of the remaining range or

$$100^{\circ}\text{F} + \left(\frac{1}{2}\right)(150^{\circ}\text{F} - 100^{\circ}\text{F}) = 125^{\circ}\text{F}$$

Arbitrarily, we choose to try 124 and 125°F (125 and 126°F would have worked just as well). The results are shown in Figure 14.10. From these results we can now say that the optimum value should lie between 100 and 125°F. We repeat this process until we can do it no more or we are satisfied with the results.

The simplest of the nonderivative oriented search procedures for multivariate functions is called *sectioning* or *one-at-a-time*. In sectioning, the searcher starts at a point and proceeds in one coordinate direction² (holding all other variables constant) in iterative steps until the value observed stops improving. The searcher then chooses another coordinate direction and repeats the process, again holding all other variables constant. This is repeated until successive searches on all N variables do not result in any improvements or the searcher is satisfied with the results. As an example, consider the two variable functions shown in Figure 14.11 in which the starting point is shown and the circular contour lines are various levels of the function. (Imagine a top view of a bowl "cut" at various levels.)

²One step in the positive direction can be tried first; if it does not improve the value, then one step in the negative direction can be attempted.

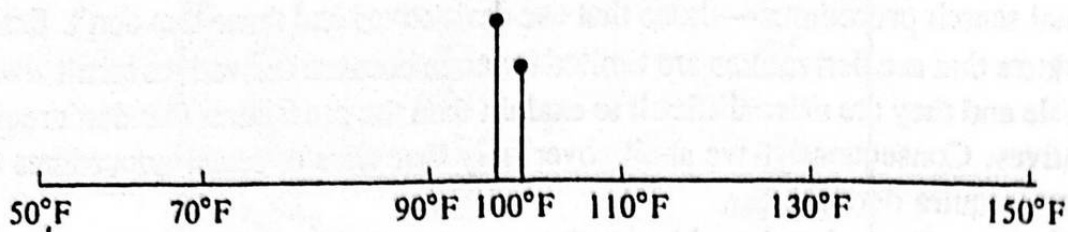


Fig. 14.9. First results of paired block search.

In this case, the searcher started at x^0 , held x_2 constant, proceeded in the positive x_1 direction until the function stopped improving at point x^1 . Then the searcher proceeded in the positive x_2 direction holding x_1 constant at x^1 and found the optimum point. Sectioning will always yield the optimum answer for perfect functions such as the one shown in Figure 14.11 in N moves, where N is the number of variables (two in this case). Usually, the function is not so well-behaved and many moves are necessary before the searcher is satisfied. Also, each move entails the evaluation of many points as small steps are taken.

As a matter of interest, the function shown in Figure 14.11 is a quadratic sum of squares with no interactive terms and equal multipliers such as

$$x_1^2 + x_2^2 \quad \text{or} \quad 2x_1^2 + 2x_2^2$$

14.8. Linear Programming

One of the first operations research methods was that of linear programming invented by George Dantzig in 1947. Linear programming is still a widely used operations research technique. Many packaged programs are available for solving linear programming problems and the basic theory is very simple, which probably accounts for its popularity.

To define linear programming (L.P.), we must go back to Equations (14.8) and (14.9) that define the general mathematical programming problem. If the objective function $f(x)$ and the constraint set functions $g_i(x)$ are all linear or first-order (the exponent of all variables equals one), then the problem is L.P. Also, since we normally deal with positive values of variables, we can add another constraint set

$$x_i \geq 0 \quad (i = 1, \dots, N) \quad (14.34)$$

Restating (to show that all functions are linear), we have the following as the L.P. formulation:

$$\text{Optimize } C'X \quad (14.35)$$

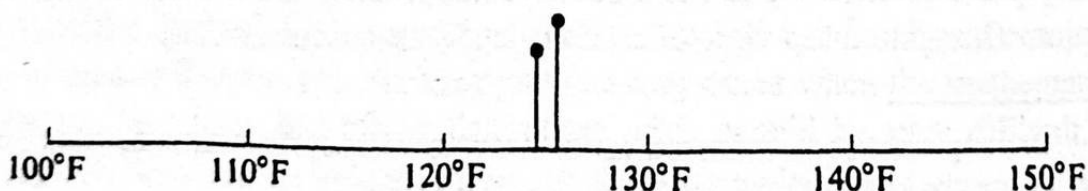


Fig. 14.10. Second results of paired block search.

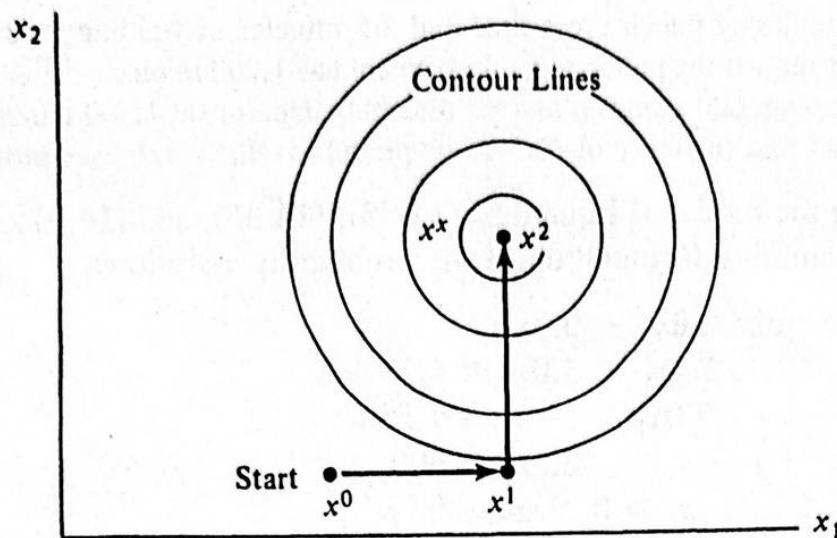


Fig. 14.11. Sectioning procedure.

subject to

$$GX \leq B \quad (14.36)$$

$$X \geq 0 \quad (14.37)$$

As illustrations, Figure 14.12 presents two linear programming problems and two problems that cannot be classified as L.P. because of the underlined elements that are not linear.

Example 14.1³

Gadgets, Inc., is trying to decide how many of each of two products are to be manufactured. A single unit of Gadgets model I requires 2.4 minutes of punch press time and 5.0 minutes of assembly time and yields a profit of \$0.80 per unit. A single unit of model

$$\begin{array}{ll} \text{Min.} & 5x_1 + 3x_2 \\ \text{S.T.} & x_1 + x_2 \leq 5 \\ & x_1 - x_2 \leq 7 \\ & x_1 \geq 0 \quad x_2 \geq 0 \end{array}$$

$$\begin{array}{ll} \text{Max.} & 3x_1 - x_2 \\ \text{S.T.} & x_1 + x_2 \leq 4 \\ & x_1 - 2x_2 \leq 1 \\ & x_1 \geq 0 \quad x_2 \geq 0 \end{array}$$

Linear Programming Problems

$$\begin{array}{ll} \text{Min.} & 5x_1 + 3x_2 \\ \text{S.T.} & \underline{x_1 x_2} \leq 5 \\ & x_1 - x_2 \leq 7 \\ & x_1 \geq 0 \quad x_2 \geq 0 \end{array}$$

$$\begin{array}{ll} \text{Max.} & 3x_1 - x_2^2 \\ \text{S.T.} & x_1 + x_2 \leq 4 \\ & \underline{\tan x_1} - 2x_2 \leq 1 \\ & x_1 \geq 0 \quad x_2 \geq 0 \end{array}$$

Nonlinear Programming Problems

Fig. 14.12. Sample mathematical programming problems.

³Partially extracted from W. J. Fabrycky, P. M. Ghare, and P. E. Torgersen, *Applied Operations Research and Management Science* (Englewood Cliffs, N.J.: Prentice Hall, 1983).

II requires 3.0 minutes of punch press time and 2.5 minutes of welding time and yields a profit of \$0.70 per unit. If the punch press department has 1,200 minutes available per week, the welding department 600 minutes, and the assembly department 1,500 minutes per week, what is the product mix (quantity of each to be produced) that maximizes profit?

Following the model of Equations (14.35), (14.36), and (14.37), we see that the linear programming formulation of the problem is as follows:

$$\begin{array}{ll} \text{Maximize } 0.8x_1 + 0.7x_2 & (14.38) \\ \text{subject to } 2.4x_1 + 3.0x_2 \leq 1,200 & (14.39) \\ & 5.0x_1 \leq 1,500 & (14.40) \\ & 2.5x_2 \leq 600 & (14.41) \\ & x_1 \geq 0 \quad x_2 \geq 0 & (14.42) \end{array}$$

Here x_1 is the number of units of model I to produce and x_2 is the number of units of model II. The objective function [Equation (14.38)] says that the total profit is to be maximized. Constraints 1, 2, and 3 [Equations (14.39), (14.40), and (14.41)] state that the punch press, assembly, and welding departments have only so many hours available per week. The last constraints [Equation (14.42)] say that only positive quantities of products can be produced.

It is a relatively simple matter to solve this problem because it is linear and there are only two variables. A two-variable problem can be drawn as shown in Figure 14.13. Each constraint is first drawn as an equation (straight line); then the appropriate side ("less than" in this case) is dashed as shown. Finally, the intersection of these areas is the "feasible region" for the problem and is shaded in Figure 14.13. For any point to be a possible solution to this problem, that point must lie within the feasible region (shaded area in the figure).

The profit for the problem is given by

$$TP = 0.8x_1 + 0.7x_2 \quad (14.43)$$

This defines a straight line for any value of profit reflecting the different combinations of x_1 and x_2 that yield that value of profit. The straight lines for profits of \$100 and \$200 are shown in Figure 14.13 by dashed lines.

It should be obvious that the total profit increases as x_1 and x_2 either move away from the origin or increase in value. Also, the TP lines are parallel. Since the solution must lie within the shaded region, the optimum point should be the point where the maximum TP line just touches the feasible region. With the slope of these TP lines, the maximum point should be point A where two of the constraints meet. Those two constraints are

$$2.4x_1 + 3.0x_2 = 1,200 \quad (14.44)$$

$$5.0x_1 = 1,500 \quad (14.45)$$

Solving these simultaneously shows that point A is located at (300, 160) and that the total profit at that point is

$$0.8(300) + 0.7(160) = \$352 \quad (14.46)$$

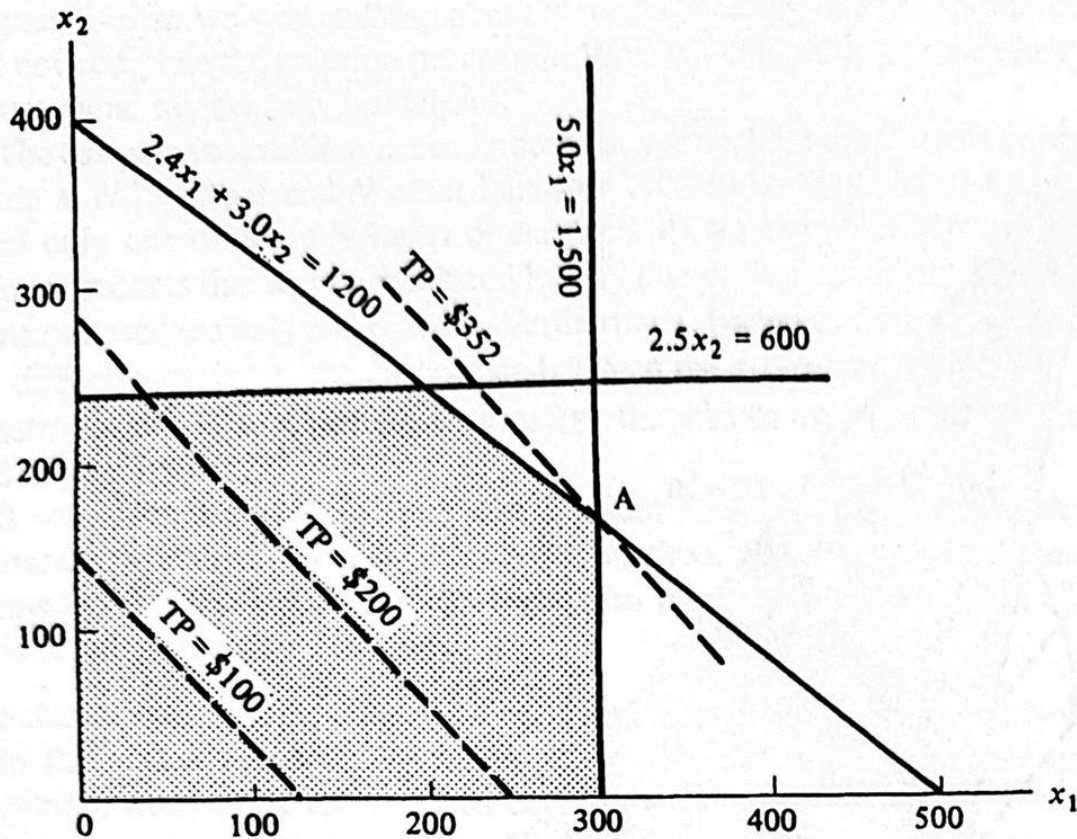


Fig. 14.13. Linear programming formulation of production problem.

That TP line is shown by the dashed line segment in Figure 14.13.

A minimization problem is very similar to a maximization problem except that the constraints are usually reversed (greater than or equal to) and it is desirable to bring the total profit line as *close* to the origin as possible. For example, Figure 14.14 represents the following linear programming problem:

$$\text{minimize total cost } TC = 5x_1 + 4x_2 \quad (14.47)$$

$$\text{subject to} \quad 5x_1 + x_2 \geq 5 \quad (14.48)$$

$$x_1 + 5x_2 \geq 5 \quad (14.49)$$

The optimum point is at $(5/6, 5/6)$, which yields a total cost of 7.5.

It is possible, but very difficult, to draw a linear programming problem of three variables (we shall not attempt one here). With three variables, the objective function is a *plane* instead of a straight line and the plane must be moved until it strikes an extreme point (edge or vertice) of the feasible region that yields an optimal answer.

Obviously, since almost all linear programming problems have more than two or three variables, how can we solve them? Since few, if any, L.P. problems are ever solved graphically, the graphical analysis of Figures 14.13 and 14.14 is good only for demonstration. Actually, almost all L.P. problems are solved by using the simplex method.

The simplex method is a mathematical procedure that systematically examines

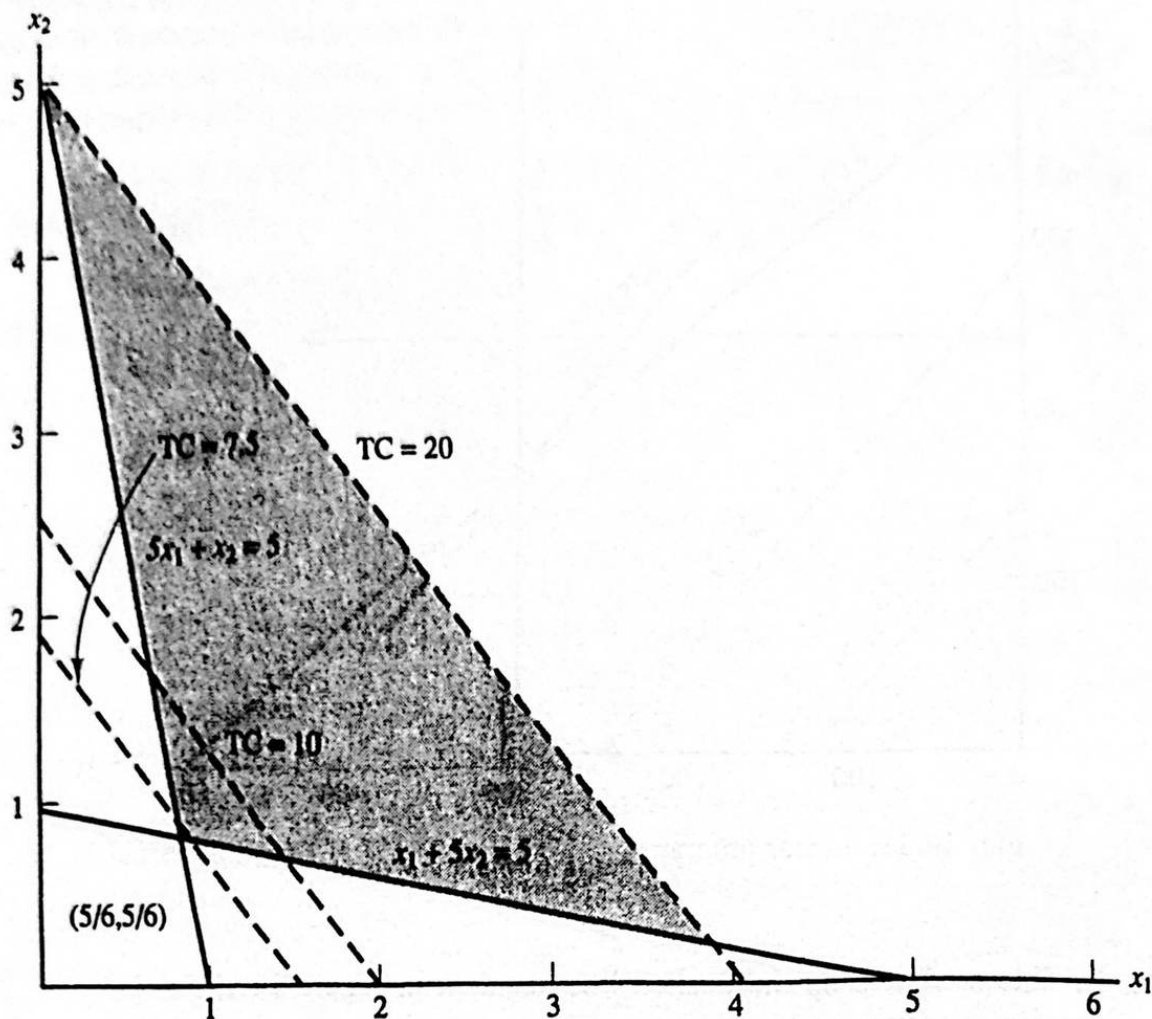


Fig. 14.14. Graphical minimization of a two-variable linear programming problem.

the extreme points of the feasible region and locates an optimum value. We can prove that at least one extreme point is an optimum for any linear programming problem; hence, the simplex method can concentrate on extreme points only. The examples of Figures 14.13 and 14.14 also *show* that the optimum must lie on a boundary, meaning that at least one optimum point is an extreme point.

Although there are many programs available for the simplex method, some people write their own programs because the simplex method is very simple to learn. The simplex method is covered in depth in industrial engineering courses and will not be covered here. Instead, let's look at some specialized linear programming problems that can be solved more easily.

14.8.1 Assignment Problem

The assignment problem is perhaps the simplest of all linear programming problems both to understand and to solve. We encountered the assignment problem

in Chapter 5 when we were talking about material handling and transportation, but we did not complete the solution procedure. We shall complete the procedure here, but, first, some review may be helpful.

The assignment problem is encountered any time there are N items or services available at N locations and N other locations (some may be similar) that require one and only one of these N items or services. As an example, suppose that we have four products that can be produced by any one of four machines, and that any machine can produce only one product. Furthermore, because of relative efficiencies on the different products, the cost varies between the assignments as shown in the cost matrix below. The objective is to assign the jobs to the machines in order to minimize total cost.

It would be possible to solve the problem above by using a mathematical programming package, but a more efficient method, the *Hungarian method*, has been developed. The first part of the Hungarian method was presented in Chapter 5 and is repeated here for convenience.

- (1) Subtract the smallest number in each row from every entry in that row.
- (2) In the matrix resulting from step 1, subtract the smallest number in each column from every entry in that column.
- (3) In the matrix resulting from step 2, a zero cost assignment (if it can be made) is optimal. If not, complete the Hungarian method.

	Machines			
	A	B	C	D
Jobs 1	16	14	15	18
2	12	13	16	14
3	14	13	11	12
4	16	18	15	17

As step 3 implies, the procedure above frequently is all that is necessary; it is impossible, however, to find a zero cost assignment for the reduced matrix shown in Figure 14.15. Here the assignment matrix has been reduced as far as possible with no zero cost assignment available.

To solve this, it is necessary to enter the next phase of the Hungarian method, which attempts to create more zeros in the matrix so that a zero cost assignment can be made. The steps of the second phase of the Hungarian method are as follows:

Fig. 14.15. Reduced assignment matrix with no zero cost assignment.

	1	2	3	4	5
1	7	3	2	0	0
2	4	0	1	1	1
3	0	0	0	1	1
4	1	1	1	3	0
5	1	1	1	6	0

- (4) Draw the minimum number of horizontal and vertical lines (no diagonals) that intersect all the zeros (four in this case as shown below).

	1	2	3	4	5
1	7	3	2	0	1
2	4	0	1	1	2
3	0	0	0	1	2
4	1	1	1	3	0
5	1	1	1	6	0

- (5) Subtract the smallest uncrossed number from every other uncrossed number and add it to all elements where the vertical and horizontal lines intersect. (The smallest uncrossed number is 1.)

	1	2	3	4	5
1	7	3	2	0	1
2	4	0	1	1	2
3	0	0	0	1	2
4	0	0	0	2	0
5	0	0	0	5	0

- (6) If a zero cost assignment can be made, it is optimal. If not, repeat steps 4 and 5 until one can be made. (In this case, a zero cost assignment can be made as shown by the circled items below. That solution is optimal.)

	1	2	3	4	5
1	7	3	2	①	1
2	4	①	1	1	2
3	0	0	①	1	2
4	①	0	0	2	0
5	0	0	0	5	①

⇒

	1	2	3	4	5
1				1	
2		1			
3			1		
4	1				
5					1

Solution

The assignment problem is not a difficult problem to solve manually even when the second phase of the Hungarian method is required. It is extremely difficult, however, to program a computer to follow this procedure and do it efficiently.

We have mentioned previously that unbalanced assignment problems (number of sources does not equal number of destinations) can be solved by adding dummy rows or columns. The procedure is very simple, for all we need do is add enough dummy sources or destinations to "balance" the problem. The costs associated with these rows or columns are all zeros to ensure that the solution is not affected.

Figure 14.16 presents two unbalanced assignment problems that have been balanced so that they can be solved by using the standard procedure.

Finally, the objective is sometimes maximizing as opposed to minimizing. In the previous example since $\max f(x)$ yields the same *solution point* as $\min -f(x)$, the change for maximization is very simple. We replace all the cost elements by their

Unbalanced						Add Dummy Row	Balanced					
	1	2	3	4	5			1	2	3	4	5
1	5	3	4	1	6		1	5	3	4	1	6
2	2	5	6	7	3		2	2	5	6	7	3
3	5	2	1	6	4		3	5	2	1	6	4
4	4	5	1	2	1	4	4	5	1	2	1	
							5	0	0	0	0	0

Unbalanced				Add Two Dummy Columns	Balanced					
	1	2	3			1	2	3	4	5
1	5	3	1		1	5	3	1	0	0
2	1	4	2		2	1	4	2	0	0
3	3	1	4		3	3	1	4	0	0
4	6	5	6	4	6	5	6	0	0	
5	3	6	2	5	3	6	2	0	0	

Fig. 14.16. Unbalanced assignment problems.

negative and continue with the minimization procedure. As an example, consider the problem regarding the assignment of salesmen to territories, shown in Figure 14.17, in which each element in the matrix shows profits and the objective is to assign the salesmen to maximize profits.

First, we must replace all elements with their negative. Then we must solve the problem by using the Hungarian method as shown in Figure 14.18. Note that only the first phase of the Hungarian method is necessary and that the solution involves assigning salesman 4 to territory 1, salesman 3 to territory 2, salesman 2 to territory 3, and salesman 1 to territory 4.

14.8.2 Transportation Problem

We have already seen an example of the use of the transportation formulation. It was presented in Chapter 4 and involved the shipment of product from M sources to N destinations.

The transportation problem is actually similar to the assignment problem in that there are M sources with something available and N destinations needing something. The difference is that under the transportation formulation, each source may have more than one unit available and each destination may need more than one unit.

The mathematical formulation is presented below for a typical minimization transportation problem:

$$\text{Minimize } E = \sum_{i=1}^M \sum_{j=1}^N e_{ij} x_{ij} \quad (14.50)$$

$$\text{subject to} \quad \sum_{i=1}^M x_{ij} = b_j \quad (j = 1, \dots, N) \quad (14.51)$$

$$\sum_{j=1}^N x_{ij} = a_i \quad (i = 1, \dots, M) \quad (14.52)$$

		Territory			
		1	2	3	4
Salesmen	1	5	3	5	4
	2	2	1	6	3
	3	5	4	6	2
	4	6	3	1	4

Entries Are In
100,000's of Dollars

Fig. 14.17. Maximization assignment problem.

where

$$\sum_{i=1}^M a_i = \sum_{j=1}^N b_j \quad (14.53)$$

e_{ij} = cost to ship 1 unit from i to j

x_{ij} = number of units shipped from source i to destination j (integers only)

a_i = number units available at i

b_j = number units needed at j

The objective equation (14.50) states that the cost of transporting the units increases linearly with the number of units such that the total cost is obtained by multiplying the cost per unit (e_{ij}) by the number of units (x_{ij}). The first set of constraints [Equation (14.51)] states that each destination needs exactly b_j units so that the sum of units shipped from all sources to the destination must be b_j . Similarly, since each source has exactly a_i units available, the total shipped out must be exactly a_i [Equation (14.52)]. Finally, the algorithm for optimally solving the transportation problem requires that the total number of units available must equal the total number needed [Equation (14.53)]. If this is not true, it can be forced by adding a dummy row or column with zero cost elements and required number of units to balance. For examples of the formulation presented in Equations (14.50) through (14.53), refer to Chapter 4.

The only solution procedure presented thus far is the least cost assignment whereby the analyst chooses the least available cost and assigns as much as possible.

		Smallest Number						Smallest Number			
		1	2	3	4			1	2	3	4
1	-5	-3	-5	-4	-5	⇒	1	0	2	0	1
2	-2	-1	-6	-3	-6		2	4	5	0	3
3	-5	-4	-6	-2	-6		3	1	2	0	4
4	-6	-3	-1	-4	-6		4	0	3	5	2
		1	2	3	4			1	2	3	4
1	0	0	0	⓪		⇒	1	0	0	0	⓪
2	4	3	⓪	2			2	4	3	⓪	2
3	1	⓪	0	3			3	1	⓪	0	3
4	⓪	1	5	1			4	⓪	1	5	1
		1	2	3	4			1	2	3	4
		0	2	0	1			0	2	0	1

Fig. 14.18. Maximization assignment problem solution.

Next, the analyst corrects the source availabilities and destination requirements and repeats the procedure until all allocations have been made. It should be intuitively obvious that this procedure does not yield an optimum answer but that it does present a very good starting solution that may be improved.

As an example, let us use the transportation problem and solution demonstrated in Figure 4.6 and repeated in Figure 14.19. Although there are other shorter ways, the easiest procedure to understand is to check each combination that is *not* currently in the solution to see if the solution is improved by including it in the solution.

For example, the Tulsa–Washington (1, 1) combination is not in the solution. If one unit were shipped from Tulsa to Washington, then a unit would have to be subtracted from the Tulsa–Lincoln (1, 3) combination or the Tulsa–San Francisco (1, 4) combination to balance the load. If one unit were subtracted from the Tulsa–Lincoln (1, 3) combination, then it would have to be added somewhere in column 3. There is no way of doing this because there are no more entries in column 3. (We cannot *add* a new one since we can prove that the transportation solution can have at *most* $M + N - 1$ combinations in the solution, 7 in this case. Therefore, the only elements that can be added to or subtracted from, *except for the one being examined*, must already be in the solution.) For example, considering the shipment of one additional unit from Tulsa to either Washington, Cleveland, or Phoenix results in cost d_{ij} as follows:

$$d_{11} = c_{11} - c_{14} + c_{34} - c_{31} = 80.00 - 78.00 + 74.50 - 70.80 = \$5.70$$

$$d_{12} = c_{12} - c_{14} + c_{34} - c_{32} = 78.00 - 78.00 + 74.50 - 72.00 = \$2.50$$

$$d_{15} = c_{15} - c_{14} + c_{24} - c_{25} = 77.00 - 78.00 + 71.50 - 70.70 = -\$0.20$$

To From	1. Washington	2. Cleveland	3. Lincoln	4. San Francisco	5. Phoenix	Plant Capacity
1. Tulsa	80.00	78.00	77.00	78.00	77.00	7,000
			4,000	3,000		
2. Tempe	76.50	75.00	73.50	71.50	70.20	5,500
				2,500	3,000	
3. Sparta	70.80	72.00	75.00	74.50	75.00	12,500
	5,000	6,000		1,500		
	5,000	6,000	4,000	7,000	3,000	

Fig. 14.19. Solution to transportation problem by using least cost assignment.

Shipping a unit from Tulsa to Phoenix (1, 5) would improve the solution by \$0.20. Since it is logical to ship as many as possible, we would have to look at the other elements affected. To add to (1, 5), we would have to subtract from (2, 5), add to (2, 4), and subtract from (1, 4). We can see, then, that we can ship 3,000 units from 1 to 5, none from 2 to 5, 5,500 from 2 to 4 (2,500 + 3,000), and none from 1 to 4, yielding the solution shown in Figure 14.20.

To check for optimality, the calculations for all the empty combinations would have to be done again. If all are positive, the solution is optimal. In this case, they are all positive. Therefore, the solution shown in Figure 14.20 is optimal.

We mentioned earlier that the transportation problem cannot have more than $M + N - 1$ positive entries in the solution matrix. It can, however, have fewer. In Figure 14.20 the solution has only $M + N - 2$ positive entries. This is okay, but it does cause some problems in trying to calculate optimality. We shall not discuss that here except to note that we might have difficulty showing that the solution in Figure 14.20 is optimal. It can be done, however.

14.9. Other Techniques

In this chapter we have only begun to skim the surface of deterministic operations research techniques. Only unconstrained optimization and linear programming have been discussed and they haven't been covered in any depth. Some of the other areas are mentioned below in just enough depth to describe them. Curricula in industrial engineering cover almost all of these techniques in detail.

14.9.1 Nonlinear Programming

We spent a large amount of time discussing linear programming because it is the simplest to understand. When the objective function and/or the constraints are

To From	1. Washington	2. Cleveland	3. Lincoln	4. San Francisco	5. Phoenix	Plant Capacity
1. Tulsa	80.00 4,000	78.00	77.00 4,000	78.00	77.00 3,000	7,000
2. Tempe	76.50	75.00	73.50	71.50 5,500	70.70	5,500
3. Sparta	70.80 5,000	72.00 6,000	75.00	74.50 1,500	75.00	12,500
	5,000	6,000	4,000	7,000	3,000	

Fig. 14.20. Revised solution to transportation problem.

nonlinear, the problem is much more difficult to solve and is called nonlinear programming. This terminology covers a large area that actually includes some of the other techniques described below. Basically, nonlinear programming includes any mathematical programming problem that does not have linear constraints and objective function.

14.9.2 Integer Programming

Sometimes a mathematical programming problem has the decision variables limited to integer values. These problems, whether linear or not, are called integer programming problems. The transportation problem is a linear integer programming problem since fractional units cannot be shipped.

14.9.3 Zero-One Programming

When an integer programming problem has the integer variables limited to the values of zero or one, it is called a zero-one programming problem. The assignment problem is an example of zero-one programming.

14.9.4 Quadratic Programming

Sometimes a problem has linear constraints but a quadratic (second-order) objective function. This is called quadratic programming.

14.9.5 Geometric Programming

Often, a mathematical programming problem has a structure such that it can be solved by using a technique called geometric programming. The structure is too complex to explain here.

14.9.6 Other Programming

Some of the other techniques encountered in the study of deterministic operations research methods are *dynamic programming*, *separable programming*, *branch and bound search*, and *mixed integer programming*. Although this list is not exhaustive, it does give an idea of the extensive field covered by deterministic operations research.

14.10. Impact of Computers

It should be obvious that almost all tools of operations research require some mathematical sophistication and a large amount of computation capability and storage space. Without the speed of computers *almost* all operations research techniques

would not be applicable to real-world problems. With the storage available on modern computers and the rapid computation time, many techniques can be used today that were only good for textbook discussion just a few years ago.

As an example, linear programming problems have been solved with many hundreds of variables and almost that many constraints. This would be completely impossible without computers. Software programs also exist for all the areas mentioned in this chapter.

Also, the unbelievable growth in power and affordability of personal computers have made larger O.R. models more feasible for the average industrial engineer. It would be rare indeed for an industrial engineer today not to have access to sufficient computing power to enable him or her to use most O.R. models in day-to-day operations. The future is practically unlimited.